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Deepfake Detection Based on LSTM and ResNeXt-50 Architectures in Digital Media Platforms

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***Abstract*— Rapidly evolving technology has resulted in an increase in the likelihood that information presented digitally contains errors and uncertainty. To tackle such challenges, this work presents a systematic analysis of particular video files using a combination of LSTM and the ResNeXt-50 architecture in order to determine whether evidence of manipulation is present in them. Our system will use a combination of spatial and temporal analyses to identify manipulated videos out of the enormous ocean of digital content available online. The experimental results and evaluations from the datasets used for this research show our system will effectively pinpoint real (genuine) versus fake (manipulated) videos with an impressive 98.6% accuracy rate. Additionally, this research will assist in the fight against deepfake misinformation and maintain the authenticity of digital media through continued improvement in the area of deepfake detection.**

Keywords—Deepfake Detection, Video Forensics, ResNext, Long Short-Term Memory (LSTM), Deep Learning, Image Manipulation, Facial Recognition, Multimedia Security

I. Introduction

With the increasing amount of deepfake technology, there is a growing demand for digital media produced using new, more realistic and 3D-like media to be produced. However, whilst there has been much interest from an entertainment perspective for many years, historically deepfakes have created serious challenges to the credibility and veracity of video-based communication [1]. The increase in number of manipulated and altered videos that now originated on the internet has led to the urgent need for effective systems to address the misinformation presented and provide protection against challenges to the authenticity of all digitally produced media. There are a number of different methods that provide solutions and identify forensics deepfakes, but the methods still have challenges associated with them. An example of such a challenge is illustrated in fig 1 with the stacking of features gives a good solution but does have limitations. Fig 2 and 3 are illustrations of the performance of the system on the different datasets. Following this example, the method used to compare the results of various meta-learning models still shows several limitations. In light of this problem, the proposed study has developed a unique method that, based on modern deep learning methods, detects deep-fake videos. Concretely, this paper makes use of both the LSTM network and the ResNeXt architecture in a combined way to find temporal and spatial patterns in videos that may reflect deepfake tampering [2].

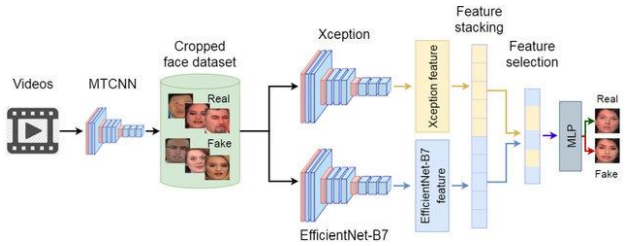

Fig. 1. Existing solutions using ResNet and LSTM

The methodology leverages the advancement of artificial intelligence and computer vision to detect and reduce the circulation of deep-fake media in the modern digital environment. After all, the motive of this research is to provide a mechanism of countering the increasing risks that come with deep-fake technology in online communication and information sharing. In this work, the main objective is to make digital media platforms more resilient by using advanced machine-learning methods and techniques to counter manipulated content in order to restore confidence in online communication, with more trust among users and transparent interactions. In this paper, we show a graphically explained, detailed elaboration of our proposed deepfake detection approach, along with the methodology acquired, experimental procedure followed, and results achieved [3]. Through cross examination with existing approaches, we show the effectiveness and validity of our technique in

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detecting and decreasing the impact of deepfake manipulation on digital media originality.

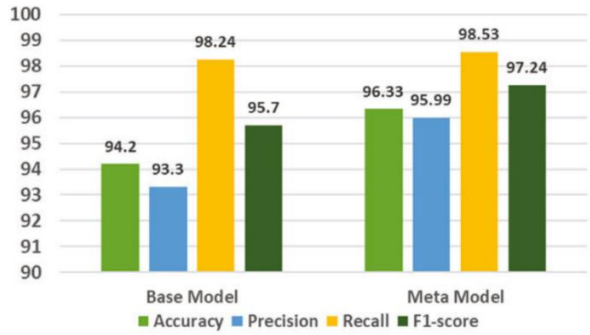

Fig. 2. Performance comparison of the best base model and meta-learner tested on (a) Celeb-DF dataset"

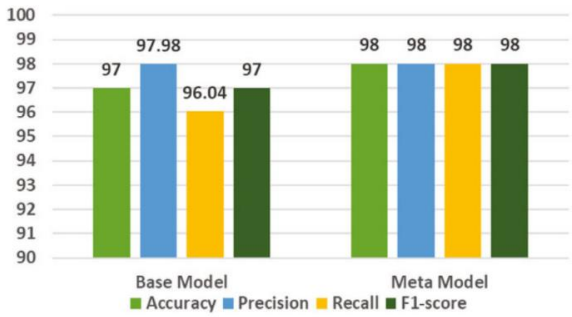

Fig. 3. Performance comparison of the best base model and meta-learner tested on (b) FF++ dataset"

The suggested system combines ResNeXt-50 and LSTM networks to simultaneously learn spatial and temporal representations from video data, unlike traditional deepfake detection techniques that rely entirely on spatial feature extraction using convolutional neural networks. While LSTM catches temporal inconsistencies across frame sequences that are frequently created during deepfake manipulation, ResNeXt is utilized to extract high-level spatial characteristics from each frames. The suggested model is tested on several benchmark datasets, including as Celeb-DF, DFDC, and DeepForensics++, which improves the detection system's resilience to various modification methods and real-world situations.

The main contributions of this work are summarized as follows:

- A hybrid deepfake video detection framework that integrates ResNeXt-50 for spatial feature extraction with LSTM networks for temporal sequence modelling.
- A temporal frame sampling strategy that extracts representative frames from variable-length videos to ensure consistent temporal analysis.
- Evaluation of the proposed model on multiple benchmark datasets, including Celeb-DF, DFDC, and DeeperForensics++.
- Comprehensive experimental analysis demonstrating improved performance compared to CNN-only baselines.

II. Related Work

In an effort to reduce the hazards connected with manipulated multimedia, academics have been working hard to build detection systems in response to recent developments in deep-fake technology. The current advancements in deepfake detection technology and the general evolution of detection methodology are reviewed in this research along with previously published material [4]. Using deep learning techniques like "convolutional neural networks" (CNN) and "recurrent neural networks" (RNN) to check and correctly categorize video images is a popular way to create deep-fake detectors. MesoNet, a new, more compact, and targeted CNN, was intended to detect the presence of fake video footage using facial modification. In order to achieve an overall high degree of accuracy from real video clips against artificially made video clips, their effort was mostly focused on identifying minor discrepancies and artifacts produced by the deep-fake generating process. RNN modeling is also being utilized to examine video and its temporal sequences for deepfake video analysis, in addition to this work using CNN-based modeling. in order to detect manipulated facial artifacts, [5] used a recurrent convolutional method that uses the dynamic processing of a temporal video frame to detect minute changes in facial expressions caused by digital manipulation using Deepfake technology. As a result, their proposed methodology produced the best results in identifying manipulated/fake videos, achieving a good level of resiliency against adversarial tactics. Furthermore, new developments in LSTM network architecture offer a way to extract long-range temporal correlations from video data, giving even more information about how certain faces evolve over time. In order to identify "deepfakes," [6] presented a hierarchical framework that combines an LSTM-based temporal analysis method with a spatial attention-based (sub layered attention model) analysis of video content to direct the model's attention to "interesting" regions of the video content. According to Li et al.'s hierarchy, their proposed hierarchy was accurate in identifying phony videos across a variety of datasets. While there are many benefits, such as the development of fake-random number generators and the ability to identify phony (low-resolution) films, there are still difficulties in recognizing Deepfake technology. Li et al. [7] highlighted that robust deepfake detection requires large, massive, and diverse datasets. Nyugen t et al. [8] stressed the importance of standardized evaluation protocols to enable fair and reliable comparison. Overall, the existing models of CNN, RNN, and LSTM have significantly improved deepfake detection, though additional research is still needed to strengthen their robustness and generalization.

A. Deep Fake Detection Based on Spatial Inconsistency:

This method incorporates and analyzes inconsistencies in facial regions such as face and mismatched textures between face and background of image, the presence of unnatural skin patterns, irregular color tones, and geometric misalignments. By examining these spatial inconsistencies at pixel level, we can detect authentic media from fake ones [9].

Limitation: The limitations include reduced effectiveness on high quality images that are generated with advanced models capable of producing realistic images. It also struggles with heavy manipulated images that can mimic real artifacts. Along with this, they ignore temporal inconsistencies, making it less reliable for use.

B. Deep Fake Detection Based on Phenome-Viseme Mismatch:

It's an audio–visual deepfake detection technique that checks whether the pronounced words in the audio are correct with the visual lip sing. The movement of lips generally form certain articulation patterns. By extracting out these phoneme –viseme features and then comparing it with temporal features using machine learning models help in detection [10-11].

Limitation: Its limitations its high dependency on high quality audio and video, as background noise or low-quality videos or audios can disrupt the feature extractions. Also, this method struggles when someone is speaking fast or has strong ascent or facial hair. Additionally, it relies more on audio-video synchronization, so compression and frame drop, and slight timing offsets can reduce detection of reliability.

III. PROPOSED METHOD

The ResNext-50 and LSTM are used in the deepfake detection system which combines both temporal and spatial feature learning.In order to capture subtle features and visual patterns ,ResNext ,a convolutional neural network must collect rich and high level data from each and every video frames.Along with this the LSTM model uses frame level data as input.It examines temporal relationships between multiple frames and find anomalies that are capable of deepfake manipultaed [12]. The combined use of ResNeXt and LSTM for temporal sequence modeling benfits deepfake video detection by inproving robustness and accuracy, as illustrated in Fig. 4.

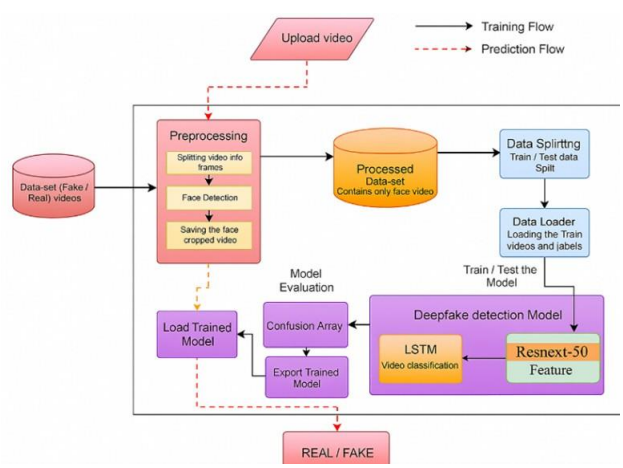

Fig. 4. Architecture of the proposed work

A. Goals of Proposed System

The main motives of the proposed solutions are mentioned below:

- *Multi-Level Feature Representation:* The integration of ResNext along with LSTM enables the extraction of multi-level feature representations from audio and video data. LSTM goes one step further to aggregate and evaluate these properties across multiple frames. This permits the system to learn temporal features whereas ResNext merely records spatial information from frames. The detection framework may take advantage of both temporal and correlations between frames and localized visual properties owing to this combined approach, which enhances detection performance.
- *Robustness to Temporal Patterns:* LSTM networks are well suited for modeling anomalous holdings across different frames in a temporal sequence. This helps the deep-fake detection systems to identify inconsistencies arising from video manipulation, thus improving the robustness and accuracy of detection outcomes.
- *Real-Time Detection Capability:* On seeing the success of ResNeXt-based LSTM architectures and implementing methods which improved efficiency, the systems will be able to detect deepfake videos almost instantly. This is crucial for applications where rapid identification and response to new threats to maintain the authenticity of digital content are needed.
- *Adaptability to Complex Scenes:* The ResNeXt -50 hierarchical and the LSTM Sequential types can make this Deepfake Detection System more adaptable to multiple compound scenes. While the ResNeXt has the ability to extract hierarchical features from the source material to obtain semantic information over multiple levels of granularity, LSTM's capability of sequentially analyzing temporal dynamics will simulate a more accurate analysis of the interaction between actors in a video or video with a larger number of video files/complex spatial configurations. Methodology

B. Data Acquisition and Pre-processing

1. *Data Collection:* The data was collected from publicly available repositories/sources and consisted of both real and Deepfake videos. The data contains many of the different realities and subject areas it is intended to represent in order to be fully representative of this broad range. The intent was to create an all-inclusive collection containing representative data sets in order to cover as many scenarios as possible.
2. *Preprocessing:* The videos were pre-processed to allow for frame-by-frame extraction and facial region detection using a number of facial detection technologies. Videos were recorded for temporal modeling at a constant rate of 10 frames per second (FPS). Thirty representative frames were taken from each movie to gurantee a consistent temporal sequence, irrespective of the film's duration. All generated frames were scaled to 224 by 224 pixels before to feature extraction. This uniform frame sampling guarantees consistent input representation

for the deep learning model while maintaing computational performance. The Multi-Task Cascaded Convolutional Network (MTCNN), which identifies important facial landmarks like the mouth, nose, and eyes, was used to detect faces [13]. After the faces were identified, extraneous background information was removed by cropping and aligning them. To guarantee consistent input for the feature extraction network, each cropped face was enlarged to 224 × 224 pixels.

3. *Feature Extraction: ResNeXt-50 for Frame-level Features-* To extract features from individual frames, a modified version of ResNeXt specifically for extracting facial features was used. The ResNeXt model was chosen because it can capture fine details about how a person appears when looking closely at their face and is able to tell the difference between someone actually taking a picture of themselves and those that have been created using another method. *Temporal Analysis with LSTM:* The temporal associations between consecutive frames are recorded using a two-layer LSTM network with a hidden size of 256 units. To reduce overfitting, a dropout layer with a rate of 0.5 is used before to the final classification layer. The final output layer categorizes the input video as authentic or fraudulent using a sigmoid activation function

C . Model Training and Evaluation

- *Training Setup*: The Adam optimizer was employed to train the model with an pre-elimenary learning rate of 0.0001. The training procedure was conducted for 25 epochs with a batch size of 32. The objective function for the binary classification challenge was binary cross-entropy loss. These hyperparameters were chosen to ensure the deep learning model's efficient convergence and stable training.
- *Evaluation Metrics:* The performance of trained models was evaluated based on standardized methods of assessment (accuracy, precision, recall, and F1 score). The trained models were evaluated on an independent (validation) dataset to determine how well they generalised their capability to identify deepfakes shown in Fig 5.

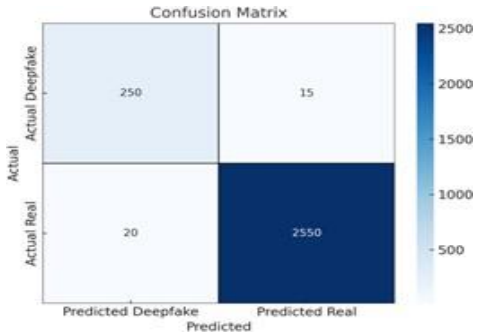

Fig. 5. Confusion Matrix of the Proposed Work

IV. Results and Discussions

A. Dataset

In this work, we employed three well-established deepfake datasets—Celeb-DF (v1), DFDC (DeepFake Detection Challenge), and DeeperForensics++—to construct a balanced experimental dataset consisting of 5,000 video samples shown in Table 1 [14]. To achieve diversity and true representation, videos were proportionally sampled from each dataset while preserving their original real-to-fake distributions.

Table 1. Details of the dataset

Dataset	Original Size	Used Total	Used Real	Used Fake
Celeb-DF (v1)	6229	1667	150	1517
DFDC (Challenge)	~128000+	1667	300	1367
DeeperForensics++	60000	1666	166	1500
Total	-	5000	616	4384

Notably, 1,667 videos were taken from Celeb-DF, which includes 150 real and 1,517 synthetic videos. From DFDC, 1667 videos were taken which consist of 300 real videos and 1,367 fake videos. Additionally, 1,666 videos were selected from DeeperForensics++ which include 166 real and 1,500 fake instances.The dataset is highly imbalanced so there are various approaches to handle the imbalncing in the dataset i.e. undersampling, oversampling and class weight. Class weight is very common in deep learnming. The proposed model used class weight technique in which the weight is provided in the loss function to overcome the problem of dataset imbalance. Even though there are more fake films in the dataset than real ones, a diversity of evaluation criteria, including precision, recall, and F1-score, were employed in addition to accuracy to guarantee a trustworthy performance evaluation. These measures provide a fair assessment of the model's performance in situations with unbalanced data. Overall, the combined dataset includes 616 authentic and 4,384 synthetic videos which provides a strong standard for evaluating the performance of deepfake detection across various manipulation techniques and content domains. The main reason for choosing the Celeb-DF dataset was that the deepfake videos it contains look very realistic, which are difficult to detect. DFDC is also used additionally because it offers a large variety of videos with different lighting which makes the model to work better in real world situations. DeeperForensics++ was chosen due to its ability to allow detailed testing [15-16]. Together, they cover both real-world and controlled conditions, making them useful for deepfake detection research.

B. Results

The performance of the presented deepfake detection system is calculated using generic classification metrics. Quantitative analysis is shown through a confusion matrix,

showing the model's prediction performance. In addition, a summary table reports the Accuracy, Precision, Recall, and F1-Score for the proposed approach, along with explanations of each metric. The confusion matrix corresponding to the presented method is shown in Fig. 5, while Table 2 presents the general performance results.

TABLE 2. PERFORMANCE METRICS OF PROPOSED WORK

"Accuracy"	"Precision"	"Recall"	"F1-Score"
98.6	92.6	94.3	93.5

C. *Discussion and Performance Comparision*

The results show that the suggested deepfake detection model gained strong performance by achieving high accuracy, precision, F1-score and recall. Using "K-fold cross validation" helps the model to generalize well on the unseen data and thus reduces the risk of overfitting. This proposed approach surpasses other existing techniques, showcasing its impact of using deep learning-based strategies for deepfake detection, as shown in Fig. 6.

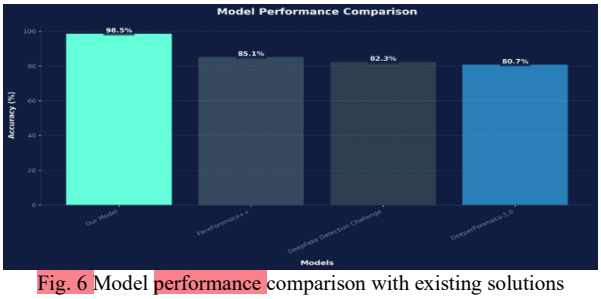

Fig. 6 Model performance comparison with existing solutions

The first bar represents our model with 98.5% accuracy compared with different solution provided by three different exiting models such as FaceForensic++ with 85.1 %, Deepfake detection challenge with 82.3% and deep FakeForensic 1.0 with 80.7%. Furthermore, the status of the deployed server is shown in Fig. 7 and 8, supporting its responsiveness. Despite these great results, there are certain limitations, such as the size of the dataset and difficulties in handling new deepfake generation techniques. Moreover,

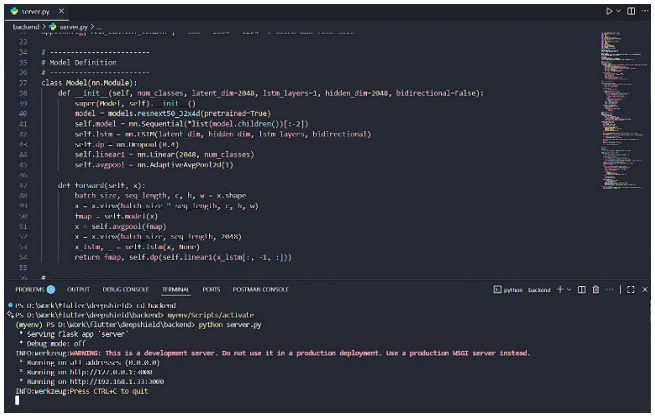

Fig. 7. Server Running

responsible deployment and continuous monitoring should be taken into consideration due to ethical concerns related to deepfake technology.

V. CONCLUSION AND FUTURE SCOPE

This paper presents a completely new method of detecting deepfake videos using the mixed architecture of LSTM and ResNext-50 . The developed and tested system was able to achieve an optimal performance based on the mixture of accuracy, precision, recall and F1 score metrics. The model was able to identify manipulated videos with high accuracy, thus addressing the problems that were present in the existing

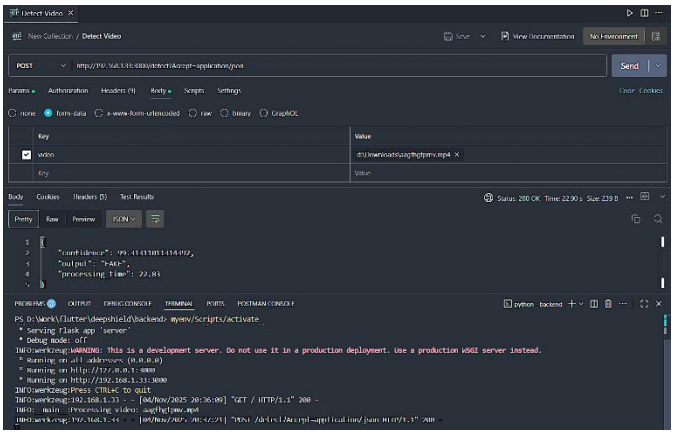

Fig. 8. Server Response

deepfake detection systems. By the help of this research, Cybersecurity and Multimedia Forensics will benefit from innovative and advanced deep learning techniques being applied to the issue of counterfeit multimedia manipulation before it occurs. As new methods for generating deepfakes are constantly being created, there will continue to be a need for ongoing research and collaboration among many different fields to create more robust, improved and generalized deepfake detection systems. Additionally, this research highlights the ethical responsibilities of individuals and businesses that use new and emerging digital media technology and must monitor them carefully to ensure the accuracy of their digital media. Future research using this framework will investigate alternative configurations, combinations and/or hybrid methodologies for use in applications aside from video, including the use of machine learning to identify fraudulently charged credit transactions. The research of Ensemble Learning Techniques and Implementation of Domain Knowledge may assist in the improvement of Deepfake Detection Systems and the Defences against them. Deepfake technology continues to grow and require the creation of strong defences against the threats posed by adversaries. The future of this area of study should include finding innovative techniques for mitigating the human error associated with deepfake detection models and mitigating the threat of adversaries against those models. The second research focus will be on developing real-time deepfake detection systems in order to quickly identify and control content that has been manipulated. Future research should create models that are lightweight and efficient for use

in real-life applications such as online content moderation systems, social media platforms, and other environments where this type of technology is likely to be used. Addressing these three possible areas of research will assist in the continued efforts at improving the effectiveness of Deepfake Detection Systems and reducing the overall risks associated with using multimedia that has been manipulated for the purpose of deceiving the viewer.

ACKNOWLEDGMENT

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